Many Daughters Pre-Analysis Plan

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In this pre-analysis plan, we outline our approach to testing each of four hypotheses (H1, H2, H3, and H4) related to the effect of having daughters on outcomes of interest. For each hypothesis, we prespecify in detail 1) a dependent variable, 2) a measure of treatment, 3) a set of covariates, 4) sample inclusion/exclusion criteria, 5) an exact analysis plan, and 6) our approach to interpreting results.

H1: Having daughters leads to stronger support for gender equality (as measured by attitudes). That is, denoting X as a measure of support for gender equality, (X for respondents with daughters) > (X for respondents without daughters).

Dependent Variable for H1 (i.e., "X" above): The dependent variable will be a standardized index of support for gender equality (std_support_gen_eq) constructed from variables *plj0620* (coded 0-10), *plj0621* (1-7), *plj0622* (1-7), *plj0623* (1-7), *plj0624* (1-7), and *plj0625* (1-7). We will use Principal Component Analysis to create an index measure based upon respondents' answers to these questions and then standardize the resulting measure to have a mean of zero and a standard deviation of one in the analysis sample. If individuals did not respond to one or two of the questions that comprise our outcome index, we will impute the mean value of that question before constructing our outcome index (and control for the number of imputed questions). If individuals did not respond to three or more of these questions, they will be dropped from the analysis sample. We will recode all variables to eliminate response codes that correspond to non-response, implausible answers, etc. (i.e., replace all negative values as missing) before constructing the index. We will code our index measure so that a larger value indicates more support for gender equality.

Measure of Treatment for H1: Our measure of "treatment" is the number of female biological children (num_daughters) to whom an individual reports being a parent. We will use the variables *kidsex01–kidsex19* to measure the number of biological children assigned to each parent.

Covariates for H1*:* We will include the following individual-level covaraites in our analytical model. We have attempted to identify the appropriate variable names for these controls in the SOEP dictionary; if we have mis-identified a variable, we will update it as part of our analysis (See "Summary of Key Variables" table below). It is our plan to categorize all inadmissible/missing questions as an "other" category.

- Individual's gender (indicators for *sex*=male, *sex*=no answer, omitted *sex*=female)
- Age at time of the survey (survey year - year of birth; *syear - gebjahr*)
- Country born in (indicators for country of birth, *corigin*)

- Degree attainment (indicators for highest level of education and indicators for whether an individual reached that level via a vocational track, an academic track, or both tracks; constructed from *pgpsbil*, *pgpbbil01*, and *pgpbbil02*)
- Employment status (full-time employed, part-time employed, not-employed, marginally employed, vocational training, other; *pgemplst*)
- Household income (gross labor income and secondary income in Euros summed over all household members; constructed from *pglabgro* and *pgsndjob*)
- Household size (the total number of people living in a household in the most recent survey year that data is available for that household; *hhgr*)
- Marital status (Single, Married, Divorced, Separated, Widowed, Registered Partnership; *pgfamstd*)
- Religion (indicator for Catholic, non-Catholic Christian, Muslim, other religion, no religion; *plh0258_h*)

We will also include the following controls:

- Indicator variables for year of survey (*syear*)
- Indicator for state/Bundesländ of residence (*hgnuts1*)
- Number of total biological* children with known sex (from *kidsex01-kidsex19*)
- Missing data indicators for all covariates or index components with missing data.

Inclusion/exclusion criteria for H1*:* We will include all parents with at least one biological child with known sex who responded to at least 4 of the following questions: *plj0620*, *plj0621*, *plj0622*, *plj0623*, *plj0624*, *plj0625*. If individuals answered one or more of these questions multiple times, we will use data from the most recent survey wave in which they were a parent with valid outcome data for each question. We will exclude responses that occurred before an individual's first child was born.

Due to the fact that child gender is more likely to be endogenous in the case of adoption or step-parenthood, we will exclude adopted or step-children from the analysis if possible (parents with both adopted/step AND biological children will be assigned their number of female biological children or total number of biological children). If it is not possible to identify whether a parent's children are biological children versus adopted or step-children, we will include all children assigned to a parent.[1,2]

[1] We note that one threat to the identification is that child gender (or: the number of female children) may not be random. One way for a family to manipulate their child's gender is to select it through a process such as invitro-fertilization. While there is evidence of preferences for sons in non-Western and developing countries (Arokiasamy, 2002) evidence on gender preferences in European countries is mixed and generally reflects preferences for balance (Karsten & Kohler, 2000). Because there is not strong evidence of a pro- or anti-female preference in European contexts and because IVF-born children represent a small fraction of all births (~2.2%, per Pfeffer, et al., 2023), we do not feel this is a major concern.

[2] Another threat to interpretation is that child gender may affect the number of children in the family. Angrist & Evans (1998) for example, have shown that family preferences for mixed-gender households can drive families with two same gender families to have a third child. As another possibility, Washington (2008) describes a "stopping" rule that could lead families to stop having children after having a child of the preferred gender, which would result in families with strong preferences having more female children in small families and fewer female children in large families. One way that we address this concern is by including family size fixed effects, so that we are comparing the incremental effect of an extra girl within families of the same size. To the extent that gender composition leads

Exact Analysis Plan for H1: We will estimate the parameter of interest (β) from the following equation:

$$Y_{it} = \alpha + \beta NumDaughters_{it} + \boldsymbol{\lambda X_{it}} + \boldsymbol{\xi_c} + \boldsymbol{\psi_s} + \boldsymbol{\gamma_t} + \epsilon_ist \quad (1)$$

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where Y_{it} is the outcome measure – here, the standardized measure of support for gender equality previously described. $NumDaughters_{it}$ is the number of biological daughters in the family (our treatment measure). $\boldsymbol{\xi_c}$ is a vector of fixed effects controlling for the total number of children *c* in the family. $\boldsymbol{\psi_s}$ is a vector of fixed effects for state of residence, and $\boldsymbol{\gamma_t}$ is a vector of survey year fixed effects. $\boldsymbol{X_{it}}$ is a vector of individual-level controls including: gender, age, country of birth, degree attainment, employment status, household income, household size, marital status, religion, and missing data indicators, as described above.

We will estimate this model using the *areg* command in Stata. We will weight responses in our regressions using the individual weight (*phrf)*, as provided by the SOEP. We will calculate confidence intervals using robust standard errors.

***Interpretation of Results for H1*:** Our test-statistic will be the z-score of the coefficient β from equation (1). If the value of the z-score of the coefficient β is greater than 1.645 (corresponding to the one-sided critical value for statistical significance at the 5% level), we will reject the null hypothesis that having an additional daughter does not increase support for gender equality and interpret this as evidence that having daughters increases support for gender equality. If the z-score is less than 1.645, we will fail to reject the null hypothesis that having an additional daughter does not increase support for gender equality

H2: Having daughters leads to more left-leaning political preferences. That is, denoting Y as a measure of left-leaning political preferences, (Y for respondents with daughters) > (Y for respondents without daughters).

Dependent Variable for H2 (i .e., "Y" above): The outcome for H2 will constructed from responses to *plh0004* (political attitude). Higher values of this variable indicate more right-leaning (less left-leaning) political preferences. We will reverse code this variable so that a higher value indicates more left-leaning political preferences and standardize the resulting variable to have a mean of zero and a standard deviation of one (std_left_lean). We will drop missing or incomplete responses.

Measure of treatment for H2t: The measure of treatment will be the same as in H2.

Covariates for H2: The covariates will be the same as in H2.

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to changes in family size, however, it is possible our estimates could reflect changes in attitudes that are related to family size, a downstream effect of child gender composition.

Inclusion/exclusion criteria for H2: We will include all parents with at least one biological child with known sex with a valid response to *plh0004*. If individuals answered *plh0004* multiple times, we will use data from the most recent survey wave in which they were a parent with valid outcome data for each question. We will exclude responses that occurred before an individual's first child was born.

Exact Analysis for H2*:* We will answer H2 by estimating equation 1 using our standardized measure of left-leaning political preferences (std_left_lean) as the outcome measure.

Interpretation of Results for H2**:** Our test statistic will be the same as H1, replacing "support for gender equality" with "left-leaning political preferences."

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household, (Z for respondents with daughters) > (Z for respondents without daughters).

Dependent Variable for H3 (i.e., "Z" above): The dependent variable will be the difference in the share of domestic work provided by the mother versus the father in a household with a heterosexual cohabiting couple with at least one child. We will measure hours of housework using *pli0043_h*, *pli0012_h*, and *pli0016_h*, and hours of childcare using *pli0044_h*, *pli0019_h*, and *pli0022_h*. We will sum hours reported for each individual in both categories and divide by the number of days for which we observe valid data for an individual to construct a measure of hours of domestic work per day per person (indiv_daily_dom_work). We will sum hours of domestic work per day per person within each couple (total_daily_dom_work) and divide each person's hours of domestic work per day by the couple's total to construct a measure of each individual's relative contribution (share_daily_dom_work=indiv_daily_dom_work/total_daily_dom_work). Our dependent variable, diff_share_dom_work will be the simple difference in this share, subtracting the value for the father from the value for the mother. Our outcome measure will thus be coded such that a higher value reflects more inequality (less equality) between genders.

Measure of Treatment for H3: The measure of treatment will be the same as in H1.

Covariates for H3: The covariates will be the same as in H1.

Inclusion/exclusion criteria for H3: For this question, we will limit our sample to hetersexual households with two co-habitating parents (one male, one female) with valid responses to *pli0043_h* and *pli0044_h*. This is because it will be necessary to have one male and one female partner to define our dependent variable. If individuals answered *pli0043_h* or *pli0044_h* multiple times, we will use data from the most recent survey wave in which they were a parent with valid outcome data for each question. We will exclude responses that occurred before individuals' first child was born.

Exact Analysis for H3: We will answer H3 by estimating equation 1 at the household level using our measure of household gender equality (diff_share_dom_work) as the outcome measure.

***Interpretation of Results for H3*:** Our test statistic will be the same as H1, but the interpretation will be reversed, since higher values of the dependent variable for H3 represent less gender equality in the household. If the z-score is less than -1.645, we will interpret this as evidence that there is a positive relationship between the number of daughters and gender equality in the household. If the z-score is greater than -1.645, we will fail to reject the null hypothesis that having a daughter does not increase gender equality in the household.

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes). That is, denoting X as a measure of support for gender equality, (X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters).

Dependent Variable for H4 (i.e., "X" above): The outcome for H4 will be the same as the outcome for H1.

Measure of Treatment for H4: The measure of treatment will be the same as in H1.

Covariates for H4: The covariates will be the same as in H1.

Inclusion/exclusion criteria for H4: We will limit the sample to respondents with known male or female gender. Otherwise, the inclusion/exclusion criteria will be the same as for H1.

Exact Analysis for H4: To answer H4, we will add an interaction between male gender and number of daughters in a family to equation (1):

$$Y_{it} = \alpha + \beta NumDaughters_{it} + \delta Male_i NumDaughters_{it} + \boldsymbol{\lambda X_{it}} + \boldsymbol{\xi_c} + \boldsymbol{\psi_s} + \boldsymbol{\gamma_t} + \epsilon_ist \quad (2)$$

Otherwise, we will estimate this equation as described in H1.

***Interpretation of Results for H4*:** Our test-statistic will be the z-score of the coefficient δ from equation (2). If the value of the z-score of the coefficient δ is greater than 1.645, we will reject the null hypothesis that the effect of having daughters on support for gender equality is not larger for males than for females. We will interpret this as evidence that the effect of having daughters on support for gender equality is more positive for males. If the z-score is less than 1.645, we will fail to reject the null hypothesis that the effect of having daughters is not larger for males than for females.

Note that the interaction term itself will not tell us if having daughters has a net negative or positive effect on the outcome: to know that, we will need to add β and δ and test if the linear combination is positive and statistically distinguishable from 0.

Summary of Key Variables

Var Name	Data Set	Content	Years	N
plj0620	pl	Gender equality index component	2016-2018	~7,400
plj0621…	pl	Gender equality index component	2016-2018	~7,400
…plj0625	pl	Gender equality index component	2016-2018	~7,400
kidsex01…	biobirth	Sex of Child #1	2015-2022 (but retrospective?)	66,316
…kidsex19	biobirth	Sex of Child #19	2015-2022 (but retrospective?)	1
gebjahr	ppathl	Year of birth	All	~1.35M
sex	ppathl	Gender	All	~1.35M
corigin	ppathl	Country born in	All	~1.35M
pgpsbil pgpbbil01 pgpbbil02	pgen	Educational attainment (secondary and tertiary)	All	773985 481139 155727
pgemplst	pgen	Employment status	All	799728
plh0258_h	pl	Religion	All	170219
pid, cid, hid	All	person/household ids	All	All
syear	All	Survey year	All	All
plh0004	pl	Political Lean	2005, 2009, 2014, 2019	90,972
pli0043_h	pl	Hours spent on housework during weekdays	All	771,531
pli0012_h	pl	Hours spent on housework during Saturday	Odd years + 2012	335832
pli0016_h	pl	Hours spent on	Odd years + 2012	408710

		housework during Sunday		
pli0044_h	pl	Hours spent on childcare during weekdays	All	771615
pli0019_h	pl	Hours spent on childcare during Saturday	Odd years + 2012	335242
pli0022_h	pl	Hours spent on childcare during Sonntag	Odd years + 2012	400619
pgfamstd	pgen	Marital Status	All except 2018	797,238
pglabgro	pgen	Gross Labor Income	All except 2018	472021
pgsndjob	pgen	Gross Secondary Income	All except 2018	30954
hhgr	hl	Current Household Size	All	528738
hgnuts1	hgen	Federal State	All	443348

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